

Aliaksei Krents

Senior AI/ML Software Engineer

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Summary

Senior AI/ML Software Engineer with 8+ years of experience developing and deploying scalable machine learning and deep learning solutions across Healthcare, FinTech, E-commerce, Cybersecurity, Automotive, Manufacturing, and SaaS industries. Proficient in Python, TensorFlow, PyTorch, SQL, and cloud platforms (AWS, Azure, GCP). Skilled in building end-to-end ML pipelines, NLP, Computer Vision, predictive analytics, recommendation systems, and MLOps (Docker, Kubernetes, CI/CD). Strong background in model optimization, data engineering, and production deployment of AI-driven applications.

Education

BACHELOR’S DEGREE in Computer Science

Belarusian Trade and Economics University of Consumer Cooperatives (BTEU)

Sep 2013 - Aug 2017

Competencies

- Large Language Models (LLMs) & NLP
- Multi-Agent Systems & Reinforcement Learning (RL)
- Computer Vision (CV)
- Predictive Analytics & Forecasting
- Model Optimization & Hyperparameter Tuning
- Cloud Platforms & MLOps
- Data Engineering & ETL Pipelines
- Real-time Analytics & Data Streaming
- Databases & Data Storage
- Microservices & Backend Systems
- Project Leadership & Cross-functional Collaboration
- AI Strategy & Innovation

Experience

Senior AI/ML Software Engineer

SoftKraft | Germany | Apr 2024 - Apr 2026

- Architected and delivered a cutting-edge autonomous ticket-to-PR engineering platform using **Python**, **FastAPI**, **Flask**, **AWS Elastic Beanstalk**, and **PostgreSQL**, enabling multi-agent LLM orchestration across **GitHub** and **JIRA** and automating 40% of low- and medium-complexity engineering tasks.
- Designed advanced multi-agent reasoning systems leveraging **PyTorch**, **Hugging Face Transformers**, **ONNX Runtime**, and **scikit-learn**, enabling contextual code generation, tool usage, and decision workflows for scalable AI integration.
- Built supervised and semi-supervised ML pipelines using **scikit-learn**, **MLflow**, **Optuna**, **TensorFlow**, and **PyTorch**, implementing automated intent classification, evaluation harnesses, and experiment tracking frameworks for better model performance.
- Engineered data ingestion and feature engineering workflows using **AWS Glue**, **Pandas**, **NumPy**, **AWS Data Pipeline**, and **SQLAlchemy**, transforming raw data and metadata into embedding-ready training datasets for model fine-tuning.
- Deployed distributed backend microservices using **Flask**, **SQLAlchemy**, **PostgreSQL**, **Redis**, and **Amazon Elastic Kubernetes Service (EKS)**, enabling efficient agent coordination, task lifecycle management, and metadata indexing across multiple systems.
- Automated CI/CD and MLOps lifecycle using **Docker**, **AWS CodePipeline**, **Kubernetes**, and **Amazon EKS**, enabling continuous training, validation, and blue-green deployment for seamless model rollouts.
- Delivered low-latency analytics dashboards using **Redis**, **WebSockets**, **Plotly Dash**, **React**, and **D3.js**, providing real-time observability and AI agent performance monitoring across production environments.

Senior AI/ML Software Engineer

Monterail | Wrocław, Poland | Nov 2021 - Mar 2024

- Led the development of an advanced **AI analytics platform** using **Django**, **Azure Virtual Machines**, **PostgreSQL**, and **Redis**, delivering scalable model-driven insights and centralized data services for cross-functional business teams.
- Built NLP intelligence layers with **TensorFlow**, **NLTK**, **spaCy**, and **OpenAI GPT-2**, enabling automated classification, real-time prediction, and knowledge extraction for enhanced decision-making.
- Designed distributed data processing systems with **Apache Spark**, **Hadoop**, **Azure Blob Storage**, and **Azure Functions**, optimizing large-scale ML training on structured and semi-structured datasets.
- Deployed containerized AI services using **Kubernetes**, **Azure Kubernetes Service (AKS)**, and **Docker**, automating CI/CD workflows with **Azure DevOps** and implementing **MLOps** for continuous model integration and validation.
- Integrated real-time analytics with **Apache Kafka**, enabling event-driven architectures and live data pipelines; built **React** dashboards for real-time model monitoring and performance tracking.

Skills

Core AI / Machine Learning

Large Language Models (LLMs)
Multi-Agent Systems & Agentic AI
Retrieval-Augmented Generation (RAG)
NLP & Conversational AI
Supervised & Semi-Supervised Learning
Model Evaluation & Experiment Tracking
Intent Classification & Task Routing
Model Drift Detection & Calibration
Feature Engineering & Dataset Curation

Programming & Frameworks

Python, FastAPI, Flask, Django, PyTorch,
TensorFlow, scikit-learn, Hugging Face
Transformers, spaCy, OpenAI APIs

Data Engineering & Processing

Pandas, NumPy, Apache Spark, AWS Glue,
Azure Data Factory, Data Ingestion
Pipelines, Structured & Semi-Structured
Data Processing, Embedding Generation &
Vector Preparation

Cloud & Infrastructure

AWS (EC2, Lambda, S3, EKS, SNS, Kinesis),
Azure (App Service, AKS, DevOps, Service
Bus), Docker, CI/CD Automation,
Kubernetes, MLOps & Model Deployment,
Autoscaling & Distributed Systems

Databases & Streaming

PostgreSQL, Redis, Elasticsearch, Apache
Kafka, Pub/Sub Architecture, Event-Driven
Microservices

Visualization & Frontend Integration

React, Plotly Dash, D3.js, WebSockets,
Real-Time Dashboards, Streaming Analytics

Project

DeepNext (Autonomous AI Software Engineering Agent)

<https://deepnext.ai>

Built entirely in the Python ecosystem and deployed on Azure, the platform leverages multi-agent LLM orchestration, machine learning pipelines, and repository-wide contextual reasoning to automate up to 40% of low- and medium-complexity development tasks.

Experience

AI/ML Software Engineer

Vilmate LLC | Warszawa, Poland | May 2019 - Nov 2021

- Built an end-to-end **AI Voice Concierge platform** using **Azure Functions**, **Azure Speech Services**, and **Azure App Service**, enabling sub-400ms voice-driven orchestration across global hospitality systems.
- Engineered **conversational AI pipelines** with **spaCy**, **TensorFlow**, **Hugging Face Transformers**, and **Azure Cognitive Services**, achieving 98% live speech-to-intent classification accuracy across multilingual environments.
- Developed predictive personalization and demand forecasting models using **XGBoost**, **LightGBM**, and **Azure Machine Learning**, powering automated routing and hospitality intelligence workflows.
- Designed large-scale feature pipelines and distributed data ingestion using **Azure Data Factory**, **Apache Kafka**, **Azure Databricks**, and **SQL Server**, processing guest profiles, booking events, and operational logs for training and inference.
- Implemented event-driven microservices architecture using **Azure Service Bus**, **Azure Logic Apps**, **Azure Event Grid**, and **Amazon SNS**, enabling real-time Pub/Sub communication between concierge services and hotel systems.
- Delivered real-time analytics infrastructure using **Azure Stream Analytics**, **Elasticsearch**, **Kibana**, **Power BI**, and **Grafana**, supporting 5M+ monthly guest interactions and providing operational insights and business intelligence.

Artificial Intelligence Engineer

Euristiq | Warszawa, Poland | Sep 2017- Apr 2019

- Architected an **agentic AI grading platform** leveraging **Django**, **Azure App Service**, and **Azure SQL Database**, delivering RAG-grounded rubric evaluation and automated **LMS synchronization** at scale.
- Built **explainable scoring pipelines** supporting essays, code, math, and oral submissions through **retrieval-augmented evaluation** frameworks using **Azure Cognitive Search**, **spaCy**, and **TensorFlow**.
- Designed **ML calibration** and **model drift detection systems** using **Azure Machine Learning**, **MLflow**, and **scikit-learn**, improving inter-rater agreement and enabling confidence-based human review routing.
- Engineered structured document ingestion pipelines transforming **PDF**, **DOCX**, **LaTeX**, **ZIP**, and **audio** inputs into evaluation-ready datasets with **Azure Blob Storage**, **Apache Tika**, and **Speech SDK**.
- Built **event-driven microservices** powered by **Redis**, **Azure Service Bus**, and **Azure Event Grid**, orchestrating grading queues, audit logging, knowledge indexing, and assignment-scoped retrieval.
- Implemented **cloud-native deployment automation** and **autoscaling infrastructure** using **Azure Kubernetes Service (AKS)**, **Azure DevOps**, and **Docker**, enabling secure, compliant, production-grade AI model serving.
- Delivered **real-time grading dashboards** and **LMS feedback synchronization** with **low-latency streaming architectures** powered by **Azure Stream Analytics**, **Power BI**, and **WebSockets**.